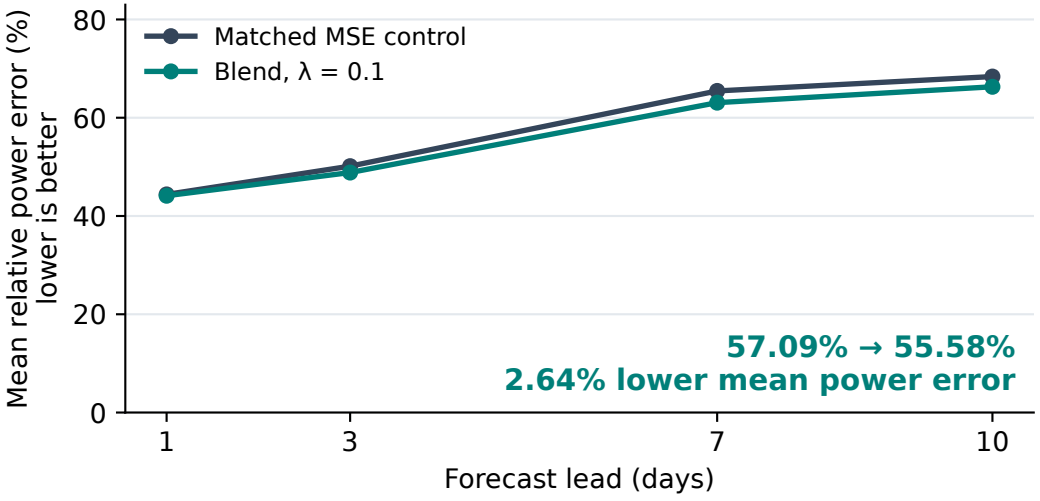
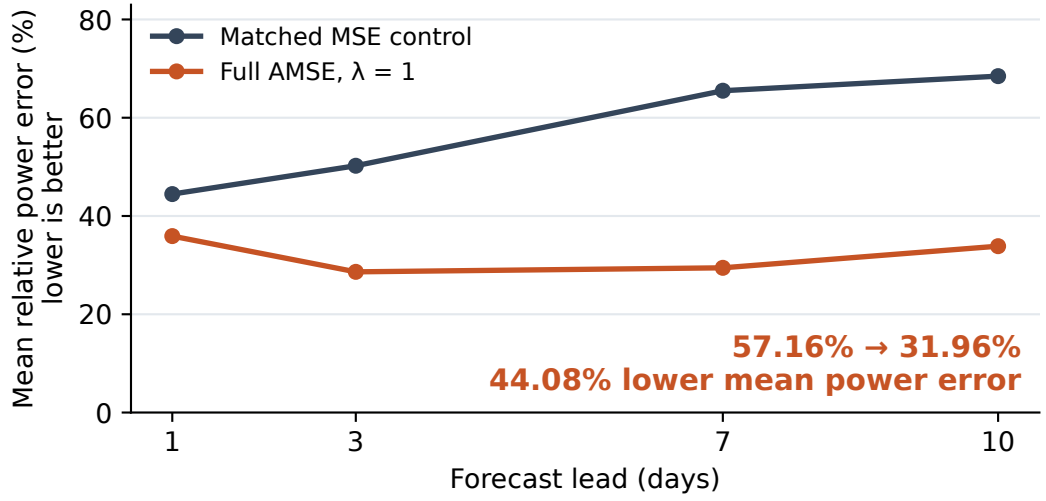
How much does AMSE improve spectral power matching?

SWA checkpoints · 8 matched 2023 starts · zonal spectra at 30–60° in both hemispheres

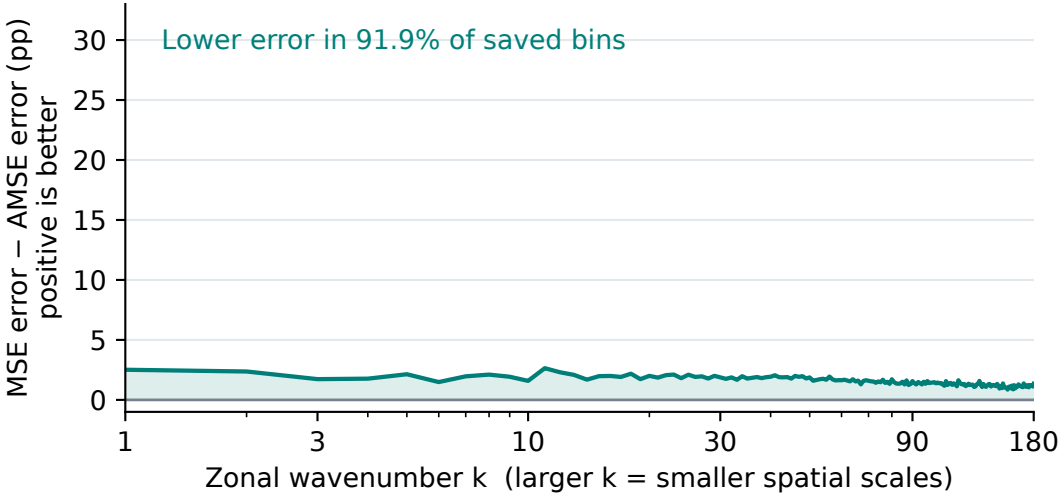
Blend, λ = 0.1



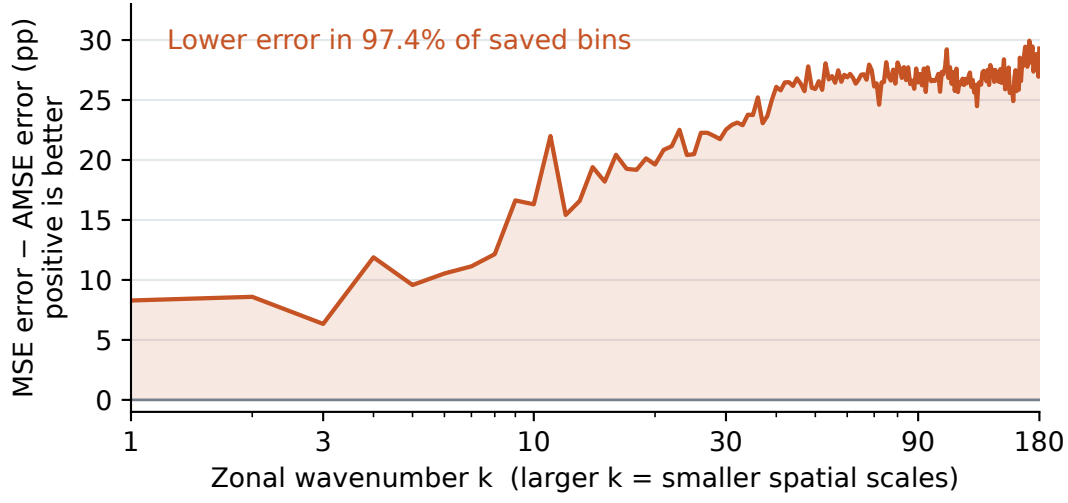
Full AMSE, λ = 1



Where power error decreases



Where power error decreases



Power error = 100 × |forecast power − ERA5 power| / ERA5 power. Raw powers are pooled across starts first.
Equal weighting of 11 fields and k = 1–180; headline and lower panels also average days 1, 3, 7, 10. No smoothing.
These are spectral diagnostics, not GraphCast loss reductions. Exploratory sample; no uncertainty intervals.